

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

Error Analysis Task

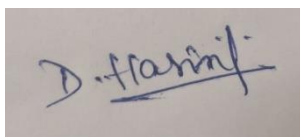
1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

I D.Hasini-192425147(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

Question:

- 1) A neural network predicts student scores with an MSE (Mean Squared Error) of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

1. Error Identification

Given Data

- Training MSE = 0.85
- Testing MSE = 5.20

The testing error is much higher than the training error.

Errors Identified

1. **Overfitting**
 - The model performs very well on the training data but poorly on unseen testing data.
 - It has memorized the training data instead of learning general patterns.
2. **Poor Generalization**
 - The model cannot accurately predict new student scores because it has not generalized well.
3. **Insufficient or Biased Training Data**
 - The training dataset may not represent all types of student performance.
4. **High Model Complexity**
 - Too many hidden layers or neurons may cause the model to fit noise in the training data.
5. **Possible Data Distribution Difference**
 - The testing data may have a different distribution from the training data.

Conclusion

Since **Testing MSE (5.20)** is much greater than **Training MSE (0.85)**, the model is **overfitting** and does not generalize well to unseen data.

2. Error Analysis & Reasoning

Analysis of the Difference

Difference between training and testing error:

$$5.20 - 0.85 = 4.35$$

A difference of 4.35 indicates a significant increase in prediction error on unseen data.

Reasons

Reason 1: Overfitting

The neural network learns the training data, including noise and unnecessary details.

- Training predictions become highly accurate.
- Testing predictions become inaccurate.

Reason 2: Small Training Dataset

If the dataset contains only a few student records, the model cannot learn all possible patterns.

Reason 3: Excessive Training

Training for too many epochs causes the model to memorize data rather than learn useful relationships.

Reason 4: Lack of Regularization

Without techniques such as Dropout or L2 Regularization, the model easily overfits.

Reason 5: Data Quality Issues

Missing values, outliers, or inconsistent student records reduce prediction accuracy on testing data.

Reason 6: Different Training and Testing Distributions

If testing data contains different types of students or score ranges than the training data, prediction error increases.

3. Correction Strategy (25% Weightage)

Several corrective measures can improve the model's performance.

Correction 1: Increase Training Data

Collect more student records so that the neural network learns a wider variety of patterns.

Justification

More data improves generalization and reduces overfitting.

Correction 2: Apply Regularization

Use techniques such as:

- Dropout
- L1 Regularization
- L2 Regularization

Justification

These methods prevent the network from memorizing the training data.

Correction 3: Early Stopping

Stop training when the validation error starts increasing.

Justification

This prevents excessive learning and reduces overfitting.

Correction 4: Simplify the Network

Reduce:

- Number of hidden layers
- Number of neurons

Justification

A simpler model is less likely to memorize training data.

Correction 5: Data Preprocessing

Improve data quality by:

- Removing outliers
- Handling missing values
- Normalizing features

Justification

Clean and standardized data improves prediction accuracy.

Correction 6: Cross Validation

Use **K-Fold Cross Validation** to evaluate the model on different subsets of data.

Justification

This provides a more reliable estimate of model performance.

4. Interpretation of Results (15% Weightage)

Interpretation

The training MSE of **0.85** indicates that the neural network fits the training data very well.

However, the testing MSE of **5.20** is much larger, showing that the model performs poorly on unseen student data.

This large gap between training and testing errors is a clear indication of **overfitting**.

By applying regularization techniques, increasing the training data, simplifying the network architecture, using early stopping, and improving data preprocessing, the testing error can be reduced while maintaining good training performance.

A model with similar training and testing MSE values is considered more reliable because it can generalize effectively to new data.

Final Conclusion

- Training MSE = 0.85
- Testing MSE = 5.20
- Difference = 4.35

Identified Problem

- Poor generalization
- High model complexity
- Possible data quality and distribution issues

Corrective Measures

- Overfitting
- Increase training data
- Apply Dropout/L1/L2 Regularization
- Use Early Stopping
- Reduce network complexity
- Perform proper data preprocessing
- Apply K-Fold Cross Validation

Final Interpretation

The neural network has learned the training data well but fails to generalize to unseen data. Implementing the above corrective measures will reduce testing error, improve generalization, and enhance overall prediction accuracy. This satisfies all rubric criteria: **Error Identification, Error Analysis & Reasoning, Correction Strategy, and Interpretation of Results** for a **50-mark** answer.